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Conceptual design of smart product-service system based on function-oriented search

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ABSTRACT

Unlike traditional products, Smart Product-Service System design requires the broad design knowledge support. However, traditional knowledge representation does not consider the characteristic elements of Smart Product-Service System, which can lead to the loss of essential information when organising and reusing design knowledge. This limitation affects the application of cross-domain functional knowledge in the conceptual design of Smart Product-Service System. To address this issue, a process model for the conceptual design of Smart Product-Service System based on function-oriented search is proposed. An initial functional modelling is first constructed based on the characteristics of the Smart Product-Service System. The extended Function Behaviour Structure model is then used to construct a design knowledge model of the system. Based on Function-Oriented Search to search for patent knowledge with highly similar functions in the patent database, the knowledge fusion is used to solve the derived problems after pruning, so as to obtain high-quality system conceptual solutions. Finally, the effectiveness of the method is verified by taking the scheme construction of intelligent forklift product service system as an example.

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KEYWORDS
Smart product-service system; function-oriented search; functional modelling; conceptual design

1. Introduction

In these past years, the integrated system consisting of products and services is referred to as Product Service System (PSS) (Hagen, Kammler, and Thomas 2018). A new type of Product-Service System, i.e. Smart Product-Service System (Smart PSS), has become the frontier of international and domestic research in the last five years. Smart PSS is based on a multi-dimensional interconnection platform, with intelligent interconnected products as the interaction interface between physical and virtual space, connecting the network of people, the network of things and the network of services, driven by data (Hara, Arai, and Shimomura 2009). As a highly integrated total solution of products and services provided by enterprises to customers, the strategic idea of Smart PSS is also gradually attracting

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the attention of enterprises (Xiang et al. 2024). A Smart PSS delivers a range of services through intelligent and interconnected products to consistently enhance the product performance for customer satisfactions (Zheng et al. 2018). The biggest difference between Smart PSS and traditional PSS is that Smart PSS adopts IT, ICT, IoT, DT and other intelligent technologies, in which real-time operation status monitoring and fault detection of the system make the information domain and physical domain of the system more closely linked. Meanwhile, industries are increasingly adopting service business models (i.e. servitisation) in order to offer not only physical products but also services as solution packages to meet individual customer needs. This convergence of digitisation and servitisation has triggered an emerging IT-driven business paradigm known as Smart PSS.

Smart PSS are able to interconnect intelligently and provide dynamically adaptive solutions, which also implies that there will be more functional coupling relationships compared to PSS, therefore PSS design methods are difficult to address the functional coupling issues of Smart PSS. Systems Engineering (SE) is a well-known methodology to assist in multidisciplinary system design, focusing on system properties (Buede and Miller 2024). The system design to validation phase can be well accomplished under the guidance of the systems engineering approach. However, for Smart PSS, it is challenging to resolve the complex functional coupling characteristics of Smart PSS and the time-varying user requirements when conflicts arise during the system engineering design process. The V-model of system engineering focuses on integration and verification activities from the early stages of system development (Zhang et al. 2023), but it lacks specific guidance on using cross-domain patent knowledge. Moreover, Smart PSS emphasises user-centred rapid iteration and continuous improvement. In the early stages of system engineering, design is equivalent to developing the logical architecture of the system (Komoto and Tomiyama 2012), by selecting and configuring different design options based on requirements, without attention to detailed design. Therefore, a conceptual design method for Smart PSS is necessary.

The conceptual design of a Smart PSS aims to ensure that products and services are tightly integrated and user-centred in the early development stages, effectively meeting the needs of the market and users while considering sustainability and long-term value. Additionally, conceptual design aids in identifying potential technical challenges and market opportunities, laying the foundation for subsequent detailed design and development. Through conceptual design, the feasibility of different options can be evaluated with less resource investment, thus increasing the probability of project success. Most current research on Smart PSS focuses on simulation and data processing analysis at the three-dimensional structure level of the product, lacking in-depth research on its functional dimensions. The object-oriented systems engineering approach enhances the abstraction and generalisation of domain knowledge in domain knowledge modelling, facilitating the sharing and reuse of such knowledge (Chao and Trappey 2024). However, object-oriented design methods focus more on the static structure of the system, making it difficult to fully capture dynamic functional interactions (Abdoli and Kara 2020). Therefore, in the conceptual design stage, the object-oriented design approach is not sufficient to address the functional coupling relationship of smart PSS in the scenarios where smart PSS requires a high degree of abstract functional analyses.

Facing this characteristic of smart PSS, this paper will design smart PSS from the functional dimension. To construct a Smart PSS, there must be a functional modelling corresponding to the whole as a general system guide to clarify the overall system framework. Therefore, it is essential to conduct modelling analysis from the functional dimension in the conceptual design stage. However, due to the multidisciplinary nature of smart product-service systems, although subsystems from different domains can be quickly designed during the actual development of intelligent products, there is a close coupling and interaction between system components across domains. This necessitates a consensus among multidisciplinary designers on the system architecture to facilitate an organic integration and form a holistic solution for the Smart PSS. However, the barriers created by the distinct bodies of knowledge in different disciplines make it challenging for multidisciplinary designers to complete system design and functional modelling based solely on their domain-specific expertise.

Therefore, in the process of Smart PSS conceptual design, designers need to collect a large amount of cross-domain innovation knowledge to solve system problems, and patents, as the world's largest source of product innovation and design information, contain 90% of the world's engineering knowledge, so patents are an important knowledge case base in conceptual design. Compared to other sources, designers can access inventions earlier in the patent literature.

Function-oriented search can efficiently retrieve cross-domain functional knowledge from patent texts to effectively support the design of Smart PSS. As a source retrieval method based on functional similarities in patent databases, function-oriented retrieval can address technical challenges in research areas by retrieving cross-domain technical solutions and knowledge transfer (Litvin 2005).

The current state of research on functionally oriented search suggests that the following problems exist in the current research:

(1) Montecchi and Russo (2015), Zhang (2010), Russo (2014) utilise various forms of functional languages to characterise the source system and design knowledge with less functional description and Functional Analysis of the target system, which hinders the effective transfer of knowledge to use.

(2) Choi et al. (2012) proposed a fact-oriented ontological approach to support FOS by using natural language processing techniques to model Subject-Action-Object (SAO) structures extracted from patent documents for fact-oriented ontology and implemented an SAO-based patent retrieval system. Fantoni et al. (2013) developed a database of patent-based product Function Behaviour Structure (FBS) models by associating subject-specific patent knowledge with FBS models using English syntactic structure rules. These studies have focused on the research of retrieval methods and the development of auxiliary tools but have yet to address their connection with knowledge representation and knowledge transfer. Moreover, there are relatively few explorations with Smart PSS as the research object.

(3) The inconsistency in the expression of cross-domain functional knowledge and difficulty in meeting the feature support of Smart PSS have brought limitations to using cross-domain functional knowledge in the conceptual design of Smart PSS.

This study aims to offer a comprehensive solution for the conceptual design of Smart PSS by starting from the actual applications and demands of enterprises and leveraging the unique characteristics of Smart PSS. The research seeks to facilitate system improvements during the early stages of the design process, thus providing a feasible overall plan for the design and study of Smart PSS. This approach addresses the challenges of integrating subsystems from various domains within Smart PSS and the inadequacies of traditional Function Oriented Searches (FOS) in providing effective knowledge support for the design of such systems. Our holistic solution aims to assist traditional manufacturing enterprises in transitioning towards more service-oriented and intelligent business models, thereby increasing their value-added services and enhancing customer satisfaction.

To achieve this, we have established a functional knowledge support framework for the conceptual design of Smart PSS based on function-oriented searching. The framework begins with abstracting the functions of Smart PSS and modelling these functions within the domains of products and services. In the process of functional modelling, we found that it can be modelled using Large Language Model (LLM), and for this reason, we specifically refined the modelling process, and in the case section, we used LLM for functional modelling of the smart forklift PSS. It then employs an extended FBS model to represent the design knowledge of Smart PSS and constructs a knowledge graph using technical knowledge extracted from patent texts. Finally, the functional modelling undergoes a refinement process through component trimming. This involves employing both component and process pruning techniques to refine the functional modelling, and using similarity calculations for knowledge retrieval to address issues that arise after pruning, thus facilitating the transfer and application of cross-domain knowledge.

The remainder of the paper is organised as follows: Section 2 presents the state-of-the-art research related to this study. Section 3 describes the conceptual design process of a Smart PSS based on functionally oriented search. Section 4 illustrates, with examples, the proposed methodology applied in the conceptual design phase to optimise the design of a Smart PSS. Finally, Section 5 concludes the paper by highlighting limitations of the study and presenting future perspectives.

2. Research status of the smart product-service system

2.1. Conceptual design of the smart product-service system

The Smart PSS comprises multifaceted functional carriers involving stakeholders, with complex combinations and interactions in different dimensions. Valencia (2017) described that the Smart PSS, as a complex and holistic solution, faces design challenges arising from two primary sources: the broad scope of Smart PSS and the dynamic nature of the Smart PSS that is constantly changing. Mirshafiee, Han, and Ahmed-Kristensen (2024) proposed a dynamic conceptual model named DHSmart and its accompanying canvas for Smart PSS. It uses visual cues to illustrate the complex interactions between the major components of a smart PSS, depicting a bottom-up transformation of the smart PSS. Cui et al. (2024) proposed a cognitive intelligence-enabled requirements elicitation framework to help designers promote the design optimisation process. Fang et al. (2024) proposed a new User-Centred Collective System Design framework and process for Smart PSS design, aiming to enhance stakeholder engagement during the entire design process, thus promoting highly effective and creative design solutions. Eisenbart, Gericke, and Blessing (2017) found that cross-disciplinary system development requires the integration of different expertise and the combination of different engineering technologies and services to provide users with expected value and desired functionality. The failure of discipline integration can lead to design errors that pose a direct threat to users and organisations. The conceptual design in the early stages of the development process can provide an appropriate means of integrating disciplines in Smart PSS. Cong et al. (2020) summarised innovative design approaches in PSS development and found that current approaches face challenges posed by the following characteristics of Smart PSS: multi-stakeholder value co-creation, closed-loop iteration of design and redesign and situational awareness in design.

Due to the multi-disciplinary nature of knowledge in Smart PSS, knowledge graphs (KGs) can effectively organise and reuse this domain knowledge. KGs were introduced by Google in 2012 with the objective of enhancing the relevance and informational content of its search engine results. By representing knowledge as a network of entities and their interrelations, KGs form a structured web, where entities and relationships are often depicted graphically (Singhal 2012). Siddharth et al. (2022) proposed a large, scalable engineering knowledge graph containing a set of $<$entity, relation, entity$>$ triples, representing real-world engineering 'facts' found within patent databases. Once effective methods are retrieved, they can be integrated with problem-solving approaches in engineering design, such as analogical design (Jiang et al. 2021). Li et al. (2020) have proposed an evolutionary design method using KG technology and the Concept-Knowledge (C-K) model. Initially, they establish two KGs using open-source knowledge, prototype specifications and user-generated textual data. Then, triggered by personalised demands, they employ four Knowledge Graph-assisted C-K operators based on graph query patterns and computational linguistics algorithms, thereby generating innovative solutions for the evolving Smart PSS. This provides a structured method of knowledge representation for Smart PSS, facilitating semantic understanding and decision support in intelligent systems by capturing and organising vast amounts of heterogeneous information.

From the studies discussed, it is evident that although research on Smart PSS has made some progress, there remains a significant research gap in areas such as interdisciplinary integration, innovation in design methodologies and systematic design approaches to address the complexities of Smart PSS. There is a need for further research to fill these gaps, particularly in developing new functional modelling and conceptual design methods for Smart PSS that can accommodate their dynamic and complex nature. Additionally, existing research often focuses on specific technologies or methods, with a relative lack of deep understanding of interdisciplinary integration and collaborative mechanisms during the design process of Smart PSS. Further exploration in this area is crucial for facilitating effective design and implementation of Smart PSS. Analysing from a functional perspective can help reveal how product services directly impact user experience, thereby providing designers with more precise guidance.

2.2. Functional modelling of the smart product-service system

There is not a complete theoretical method for the functional modelling of Smart PSS. Research scholars have conducted relevant research on the functional abstraction of Smart PSS. The system complexity and functional coupling of Smart PSS bring significant challenges to designers. Functional modelling can guide researchers to build a consensus basis for the collaborative analysis and optimisation of Smart PSS.

In terms of developing functional modelling, the methods mainly include relational functional modelling and process functional modelling (Dong et al. 2019). The TRIZ functional modelling method (Altshuller 1999) is a classic method of relational functional modelling, which expresses the functional network composed of the interaction relationship among system components, objects, supra-system components and so forth. However, it cannot express the sequence of the functional realisation and temporal change of the system state. The terminology of specific components hinders the communication and understanding of designers from different disciplines. The functional structure modelling method (Pahl et al. 2007) is a classic approach to construct a process-oriented functional modelling. This method deconstructs the overall system function into sub-functions and further into functional elements, which articulates the transformation process involving the flow of energy, materials and information. It elucidates the journey of system function realisation through the input/output relationships among these functional elements and their interrelationships. Ultimately, it leverages the input – output relationships and their associated flows to depict the process of functional realisation within the system. The abstraction of functions in Smart PSS serves as a vital cornerstone, fostering a common ground for multidisciplinary designers and catalysing seamless interdisciplinary collaboration in the systematic design process. Yan et al. (2017) investigated the synergistic mechanism of function, behaviour and structure based on the FBS model and established a conceptual design process model for multidisciplinary collaboration. Chang et al. (2019) applied the benefit–cost-expectation model and the functional structure of the product to propose a user-centred design methodology for Smart PSS based on the three dimensions of material flow, data flow and value flow. However, this method relies on designers to directly abstract specific structures into flows and their conversion relationships. While it facilitates cross-disciplinary communication, it places a heavier cognitive burden on designers during the modelling process, potentially diminishing its practical feasibility (Mokhtarian, Coatanéa, and Paris 2017). From the above scholars' studies, existing functional modelling methods are only partially applicable to Smart PSS, and when developing functional modelling, the inherent complexity of Smart PSS and the tight coupling between functions can exacerbate cognitive burdens and interdisciplinary communication barriers for designers.

In terms of populating the knowledge base, notable advancements in functional modelling research have been witnessed recently, building upon classical functional modelling methods. Chen and Xie (2017) pioneered the development of a functional knowledge integration system. This system seeks appropriate functional units in a distributed resource environment, assembles them to construct a chain of functional units, and meticulously selects the most optimal entities for these functional units. Gill, Summers, and Sen (2019) identified three distinct approaches employed by different designers in functional structure modelling: reverse modelling along outputs to inputs, forward modelling along inputs to outputs and backward and forward extension modelling centred around core functional elements. Their research has discerned that reverse modelling exhibits superior capabilities in supporting scenario-based reasoning. Kang and Tang (2014) devised an input flow similarity matrix, output flow similarity matrix and function similarity matrix by juxtaposing functional structure models of different products. They expanded their method by transitioning from single-function products to multi-function products, facilitating comparative analysis. Furthermore, Bo et al. (2014) introduced a conceptual design approach that extends product structure by using the functional unit as its foundation. This method involves constructing a functional chain diagram for design activities and subsequently establishing a functional unit – structural unit association matrix. However, it can be seen from the above scholars' studies that populating the knowledge base requires designers to have advanced knowledge management and integration skills in order to deal with a large amount of functional knowledge. This creates a certain design capability barrier for designers.

In terms of determining functional priority, various approaches have been proposed by researchers: Yu et al. (2013) constructed a multifactor evaluation system for trimming priority determination based on the system functional modelling. They combined the level of idealisation and functional value assessment to determine the order of trimming of the functional components. Domb and Kling (2006) determined the trimming components based on functional analysis to meet different design requirements. However, the trimming execution operation still exhibits a certain level of blindness. Zhang et al. (2017) proposed a systematic feature recommendation process to support data-driven feature planning in product and service design. Gui and Chen (2021) proposed a scenario integration method in intelligent system functional design for interactions of intelligent systems and their environments based on scenarios. They explicitly represented the complex functional logic of the system. Lee, Chen, and Trappey (2019) proposed a structured Smart PSS design method based on the Theory of Inventive Problem Solving (TIPS or TRIZ) and service blueprints. They constructed a conceptualised model of Smart PSS based on analysing stakeholders, product characteristics and service scenarios. Wu et al. (2021) proposed a set of methods to optimise innovative PSSs from a functional perspective based on the digital twin multidimensional framework. They aim to optimise Smart PSS scenarios in the conceptual design stage. Therefore, differences in methodological approaches to functional structure modelling and scenario-based reasoning capabilities among designers can lead to significant discrepancies when determining function priorities.

In summary, it is imperative to build upon existing research on functional modelling methods to further explore and develop approaches that are finely tuned to the unique characteristics and complexities of Smart PSS. Such efforts will play a crucial role in advancing the field and meeting the evolving demands of increasingly complex Smart PSS.

3. Design process of the smart product-service system based on function-oriented search

Based on the existing FOS methods, this paper presents a framework for managing and employing design knowledge for FOS to assist designers in acquiring high-quality design solutions for Smart PSS. Function-oriented search consists of requirement identification, patent knowledge extraction, and conflict identification. The specific process is illustrated in Figure 1. It consists of three main stages:

Stage 1: Establishing a functional modelling of the critical objectives or issues, i.e. establishing a functional modelling of the Smart PSS.

Establishing a functional modelling of a Smart PSS is the initial stage of an extended FOS. Function is a critical concept in design engineering. Researchers can be guided to build a consensus for collaborative analysis based on functional modelling. At the same time, it also lays a foundation for the application of the theory of FOS. Therefore, the premise

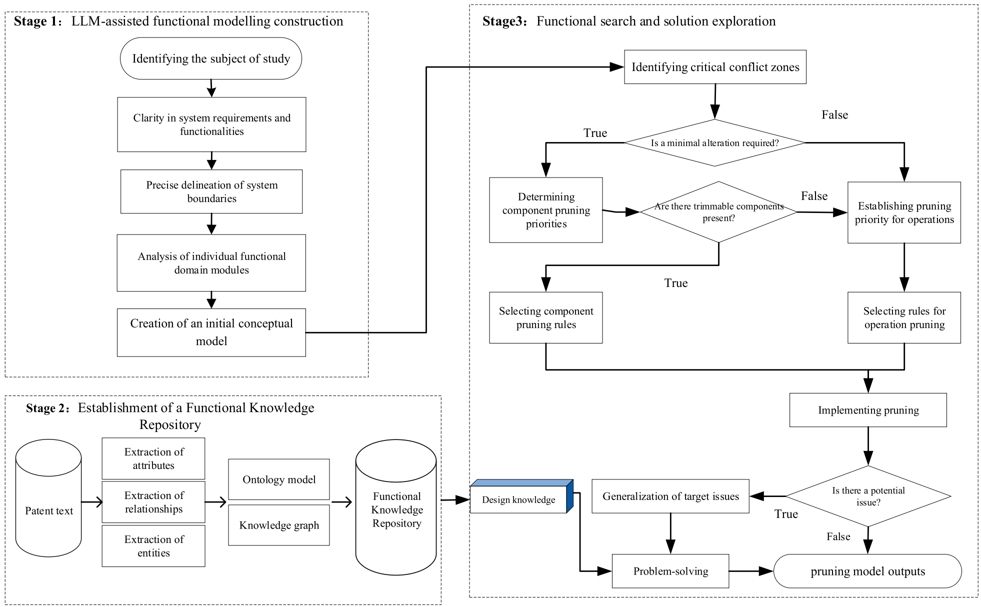

Figure 1. Conceptual Design Process Model of the Smart PSS.

of extended FOS is to complete the functional modelling of the target object, i.e. the Smart PSS.

Stage 2: Functional knowledge base construction, i.e. constructing a knowledge base to facilitate the effective management of knowledge based on a knowledge representation method for the design knowledge of the Smart PSS.

Building a knowledge base will provide a cross-domain knowledge source for designing a Smart PSS. In addition to the knowledge existing in different carriers, such as engineering libraries, manuals and essential databases, the design process of the Smart PSS requires the support of cross-domain technologies such as patent texts. The corresponding organisation and expression of diverse knowledge in the design process of Smart PSS are essential means to achieve design knowledge reuse and improve design efficiency. Therefore, in this stage, design knowledge is characterised based on exploring the ontology model of Smart PSS. Simultaneously, the establishment of a knowledge base for Smart PSS design is undertaken to facilitate the realisation of knowledge reuse and search.

Stage 3: Functional pruning (Caldwell and Mocko 2012) and design knowledge fusion involve the use of effective search methods and models to find design knowledge. Knowledge fusion is then used to transfer the knowledge gained from the search into the design of the pruned Smart PSS, resulting in the generation of new solutions.

The ultimate goal of FOS is to transfer the design knowledge obtained from cross-domain search to design a Smart PSS through knowledge fusion and to form the final conceptual design scheme based on the target design requirements. Functional modelling is built based on the requirements of the Smart PSS to identify the main conflicts as well as the cropping objects. When functional pruning is not enough to meet the design requirements, we need to use the functional knowledge base constructed by extracting patent knowledge as the analysis object to find a suitable design knowledge source based on

FOS. Finally, the technical solutions and functional principles in the design knowledge are reasonably transferred to the Smart PSS design.

3.1. Functional modelling of the smart product-service system

The realisation of Smart PSS functions relies on the close cooperation between product and service domains. Functional modelling of Smart PSS covers multiple dimensions of information physical space, composed of functional carriers such as physical products, services, software, APP, and data platforms, and contains extensive modelling elements of a multifunctional domain. Zheng et al. (2019) provided an overview of Smart PSS from the product-service level, system level, and system-system level. Wu et al. (2020) categorised smart PSS into Smart PSS, stakeholders, and intelligent service systems. Building on the work of these researchers, we considered multidimensional functional domains from three perspectives: physical entities, software, and services. However, these three parts are not isolated but interconnected, collectively forming the smart product-service system. Recently the technology of using LLM (Wang et al. 2023) to assist designers in conceptual design is emerging, by designing the following specific steps in order to provide the process of deriving the thinking of the LLM, which facilitates the establishment of a good foundation for building a better prompting process in the future. The prompts are designed as 'Role played by the LLM" + 'Task in charge," which enhances the quality of the content output by the LLM.

Step 1: Define the System Requirements.

A smart PSS is an integrated package of technological products and services, with system requirements based on the expected needs of customers or contractors. We send these expected requirements to the LLM and use design hints to let the big model play different roles to extract the design and functional requirements of the system. As shown in Figure 2.

Step 2: Clarification of System Functions.

Functions of the Smart PSS come from needs of users, and the design process of the Smart PSS requires the completion of the mapping between user needs and product functions. That is the functions are expressed in a uniform and standardised way, and the semantics of a single function is specified in a machine-understandable way so that it can be analysed or decomposed into more specific sub-functions.

Step 3: Clarification of System Boundaries.

The functions of Smart PSS are derived from user needs, and its design process involves mapping these needs to product functions. This ensures that functions are expressed in a uniform and standardised way, with each function's semantics specified in a machine-understandable manner, facilitating analysis or decomposition into more specific sub-functions. For example, if a floor sweeping robot needs to implement automatic brush head changing based on stains, then subfunctions such as stain recognition, automatic brush head changing and navigation path planning must be implemented. To realise the

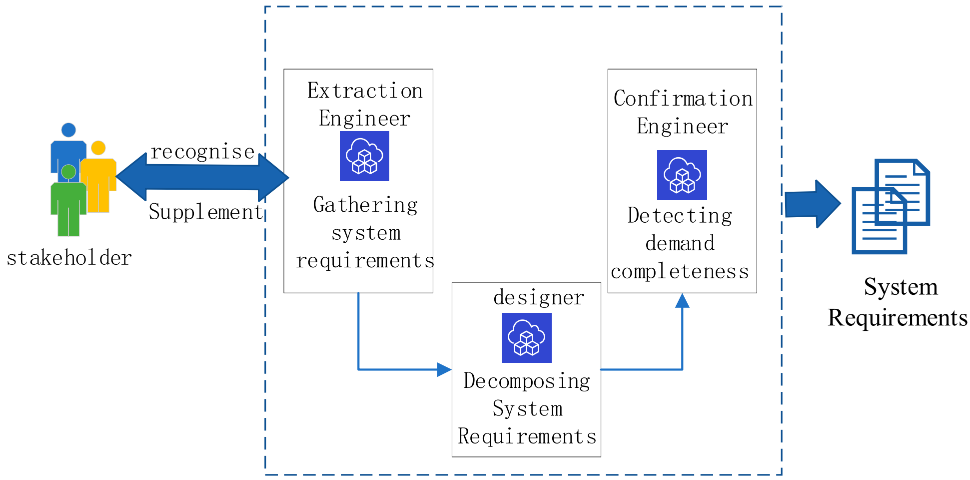

Figure 2. LLM Assistive Intelligence PSS Requirements Definition.

automatic brush head replacement sub-function, the two functional elements of automatic brush head removal and automatic brush head installation must be realised. In this way, all the functions of Smart PSS are decomposed into functional elements, thus constructing a functional tree.

Step 4: Multidimensional Functional Domain Modelling.

When viewing Smart PSS as three modules – physical entities, software and services – each module is analysed and described using a corresponding symbolic system. The symbols for system function modelling are shown in Table 1. The specific process unfolds as follows:

(1) Physical Entity Module Functional modelling

① Component analysis of the entity module, primarily focusing on three aspects. First, Clarifying the usage environment of intelligent products and analysing the specificity of their operating environment. Second, Defining the execution method of intelligent product functions, drawing from similar products on the market to enhance function execution. Finally, Determining the specific data collection method of intelligent products.

② Analysis of component properties and interaction relationships.

After identifying the components of Smart PSS modules, elements of the physical entity module are analysed and described based on component relationships and functions.

③ Diagrammatic representation of the physical entity module.

Uses the functional modelling to diagrammatically represent the physical entity module, checking for any omissions based on functional realisation effects.

Table 1. List of system functional modelling symbols.

Icon	Hidden meaning
Physical components	Indicates a specific structural element
Subordinate attributes	Indicates a description of the characteristic attributes or states of the entity
Behavioral entities	Represent the changes that occur in the entity, the operational behavioural actions and the service realisation process
Virtual components	Represent the specific non-physical entity meta-components
Stakeholders	Indicate participants in process activities in the system
Material flow Energy flow Sign Control	Represent the input/output material, energy and signal flow

(2) Software Module Functional modelling

① Clarification of software function positioning.

The role of the software within Smart PSS should be clearly defined. Based on its assigned functions, the software can be divided into two types: one that implements software functions determined during the functional decomposition process, mainly advanced control planning algorithms, designed by machine learning and control professionals; and another that coordinates the execution functions across physical, service and software modules.

② Analysis of software function execution sequence.

Designing software modules should consider their interaction with physical entity modules and service modules, potentially leading to updates in the software architecture. Any introduced design changes or constraints should be reflected in the functional modelling of the Smart PSS.

③ Diagrammatic expression of software module.

Once the elements within the software function domain are determined, the functional modelling can be employed to diagrammatically represent the conceptual schemes of the software modules in Smart PSS.

(3) Service Module Functional modelling

The service module includes general services provided by staff and mobile services that operate online and offer value-added solutions to customers. The provision of services requires the participation of both service providers and service users.

① Analyse roles of different stakeholders in the system.

Designers can identify available personalised services based on specific user scenarios, clarifying potential user behaviours in different contexts.

② Clarify the interaction patterns of different stakeholders with the service module.
③ Diagrammatic expression of service module.

3.2. Design knowledge management for function-oriented search

3.2.1. Knowledge representation of the Smart PSS design based on knowledge graphs

The considerable, multi-source and interdisciplinary engineering knowledge resources contribute to the development of Smart PSS. To support the knowledge-intensive process in Smart PSS development, effectively organising and reusing this domain knowledge is essential. Addressing this issue, a knowledge representation model for innovative Smart PSS based on a knowledge graph is proposed. This model represents attributes and relationships among multiple entities in design knowledge.

In order to establish a knowledge management system, an effective knowledge representation is indispensable. KGs offer high extensibility and interpretability, particularly for linguistic forms of knowledge, making them practical tools for supporting knowledge representation. Design knowledge contained within a patent can be expressed as a design knowledge graph or as part of a graph: $K = \ < P, R, S, C >$, where P, S and C represent entities or nodes of design knowledge in the graph, while R represents the direct semantic relations between P and S, as well as between P and C. Therefore, the design knowledge base of a Smart PSS consists of a series of independent design KGs: Knowledge_Base $=$ K1, K2, K3, … , Kn.

The ontology model offers a formal and shared semantic representation, enhancing the efficiency of knowledge representation. The Smart PSS design ontology model depicts relationships and attributes between various conceptual entities in product or service modules in an abstract manner. Through a PSS design ontology model, the definition of design knowledge is guided by references, reducing ambiguities arising from varying representations among different knowledge sources.

The Smart PSS design ontology abstracts and defines the Smart PSS design conceptually, covering the system's functions, service activities, system structure and usage scenarios.

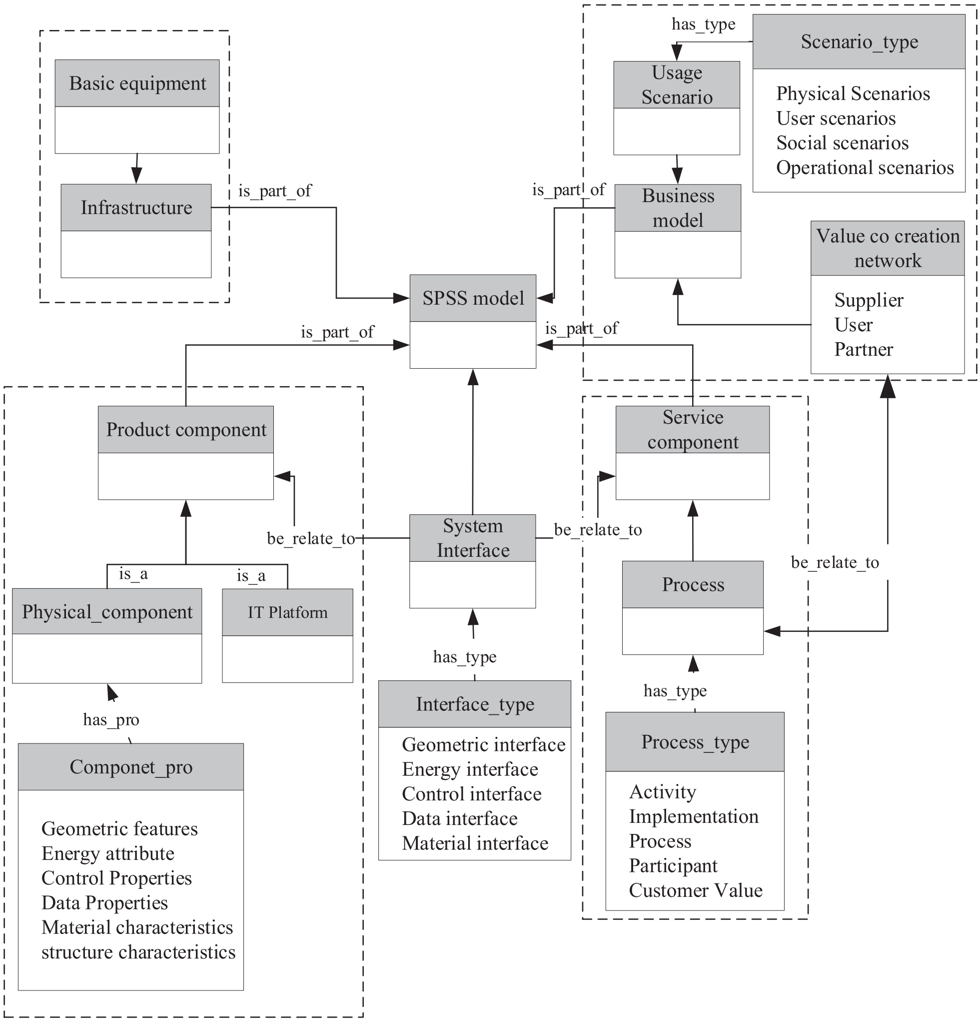

Figure 3. Smart PSS Design Ontology Model.

As shown in Figure 3, the Smart PSS Design Ontology offers a semantic description of the design case conceptually, providing further guidance for the construction of the design case.

In order to model the components/modules and their relationships in a Smart PSS across multiple usage scenarios, this paper employs the FBS ontology during the design phase to construct the knowledge graph. By expanding upon the traditional FBS model, behavioural and structural information is organised in relation to both products and services, as shown in Figure 4.

The essence of constructing a knowledge representation model lies in using the extended FBS model to describe the design knowledge of Smart PSS. To model the design knowledge in patents and characterise its distinguishing elements, this paper proposes a specific modelling process for design knowledge mapping of Smart PSS, outlined in the following steps:

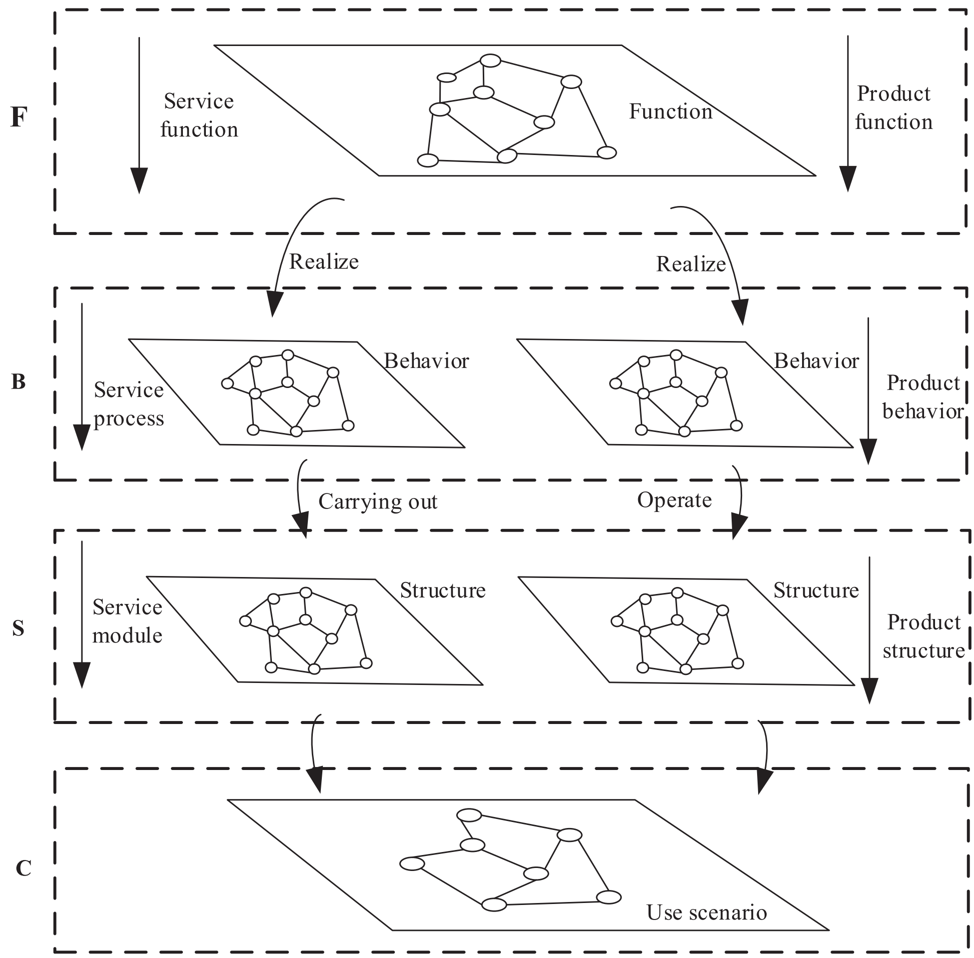

Figure 4. FBS Model Extension.

Step 1: Use requirements analysis to define the main functions of the patent design.

Step 2: Analyse the main behavioural entities that implement the central function realisation.

Step 3: Analyse the attribute characteristics of the primary behavioural entities and related flows, expanding the ontology as needed based on actual requirements.

Step 4: Analyse the constituent elements of the patent design based on the main behavioural entities, determining the primary structure of the constituent system.

Step 5: Define the analysed structural, behavioural and other entities as graph nodes, establishing input/output relationships between different entities.

Step 6: Identify key attributes of each node involved in realising the functional role in the graph.

Step 7: Incorporate information on the usage scenarios of the patented design.

Step 8: Verify the completeness of the knowledge graph.

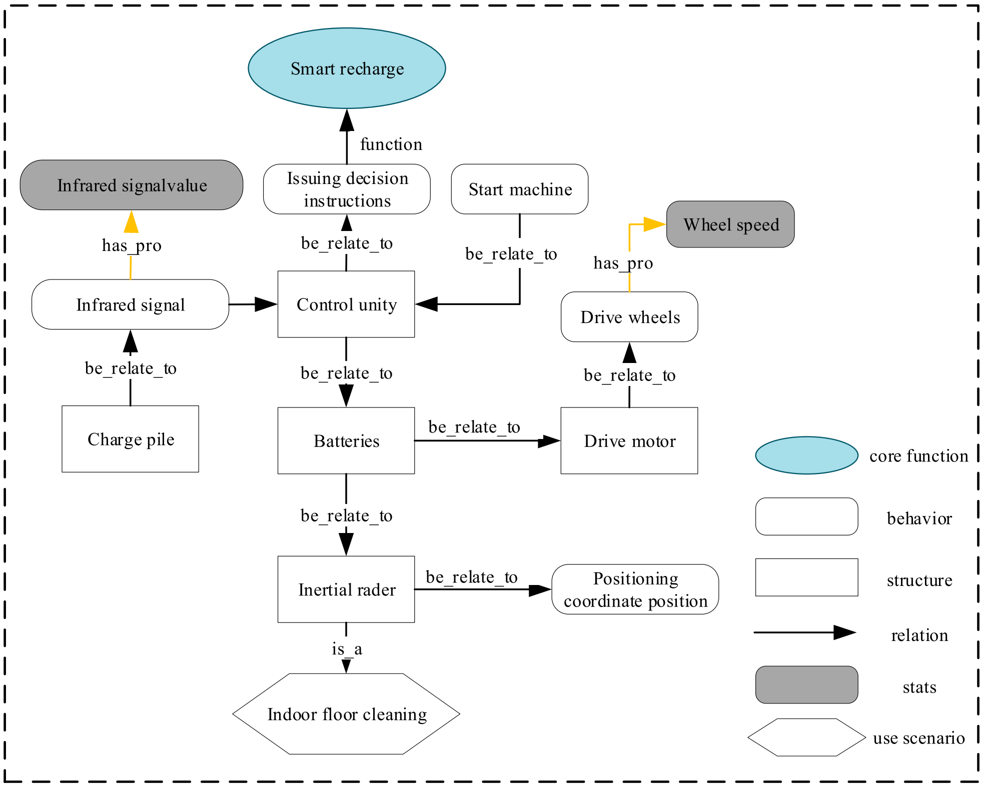

Figure 5. Example of a Knowledge Map.

As shown in Figure 5, this paper uses an invention patent, specifically the Intelligent Fast Return Seat Method (CN111543897A) of sweeping robots, as an example for constructing the knowledge graph (Xufen 2021). During a sweeping task, when the sweeping robot detects the infrared signal of the return seat via the control module, it assesses whether it should enter the return mode. If the current infrared signal value surpasses the original maximum, the control module directs the sweeping robot to initiate the return docking programme.

3.2.2. Functional knowledge base construction

Storing functional design knowledge extracted from patent texts can serve as a valuable knowledge source for Smart PSS design. This study presents the development of a prototype system called the SPSS support system, which is a knowledge-based design support system for Smart PSS. The SPSS support system is built on the Windows 10 operating system using Python's pyqt5 framework as the application development tool, adhering to the principle of separating business and logic. Data management is handled using the MySQL database to organise multidisciplinary functional knowledge in a format akin to the design catalogue. Data retrieval from the database involves conversion into data frame format and subsequent presentation in the Tab-view control. The prototype of the Smart PSS Design Support System software is shown in Figure 6.

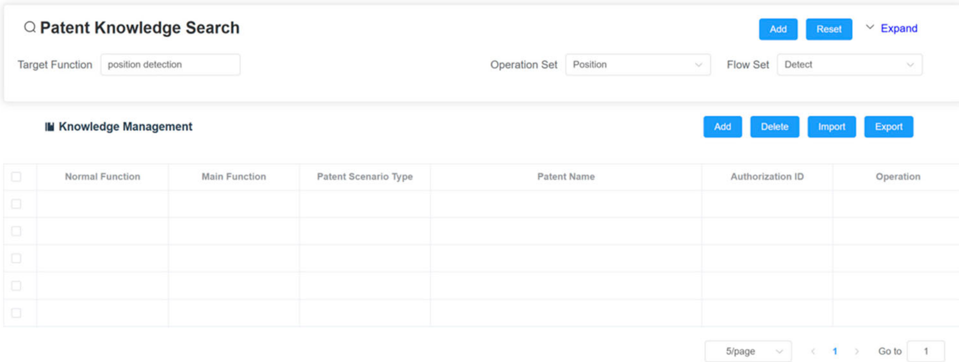

Figure 6. Design Knowledge Search Interface.

3.3. Design process based on functional pruning and knowledge support

In addition to employing effective functional knowledge search methods, integrating design knowledge into the Smart PSS is crucial for enhancing the solution quality.

3.3.1. Analysis of conceptual solutions based on functional pruning

Smart PSS integrates complex components into a unified system with intricately coupled functional characteristics. Pruning, an essential method for optimising Smart PSS design, involves reducing complexity and enhancing idealist by eliminating problematic functional elements from the system's functional modelling. In addition, pruning allows for the optimisation of the system structure using either internal or external resources. Ultimately, the optimised product can better adapt the Smart PSS to new market demands. Leveraging the multidimensional functional domain model obtained from the four steps outlined in Section 3.1, we proceed to the next step.

Step 5: Determine the Pruning Object.

A Smart PSS contains different devices, making the function realisation process complex. Direct analysis of the entire system involves numerous factors, rendering the analysis process complicated. To effectively enhance the efficiency of system analysis, we propose rules for determining cropping priority for Smart PSS and selecting corresponding rules based on enterprise needs. The specific methods are outlined as follows:

Rule 1: Analyse low paths of defect streams in the conceptual scheme of the Smart PSS to identify critical conflict areas.

If the function in the conceptual scheme of the Smart PSS is inadequate or detrimental, resulting in output failure to meet user goals, it indicates a problem area in the defective flow. This analysis helps quickly identify key conflict areas and enhances the efficiency of Smart PSS process analysis.

Table 2. Triangular fuzzy number scale and meaning.

Scale	Triangular fuzzy number	Hidden meaning
0.9	(0.8, 0.9, 0.9)	particular importance
0.8	(0.7, 0.8, 0.9)	very important
0.7	(0.6, 0.7, 0.8)	general importance
0.6	(0.5, 0.6, 0.7)	slightly important
0.5	(0.4, 0.5, 0.6)	equal importance
0.4	(0.3, 0.4, 0.5)	inverse comparison
0.3	(0.2, 0.3, 0.4)	
0.2	(0.1, 0.2, 0.3)	
0.1	(0.1, 0.1, 0.2)	

Rule 2: After determining the critical conflict area of the Smart PSS, the enterprise or designer selects process pruning or component pruning based on actual requirements.

If a company cannot resolve a problem through component pruning or seeks higher innovation, it can tailor critical problematic operations in the Smart PSS to compensate for missing useful functions by leveraging practical resources.

Rule 3: Analyse the path of action generating the defect flow and select vital problematic elements or functions to be pruned based on functional value analysis of elements within the critical area.

If an enterprise has limited capabilities or prioritises higher reliability, it may opt to minimise system changes. By identifying minor issues in the actual problem situation of the Smart PSS, minimal modifications can be made while striving to minimise costs for system improvement. For this type of business, critical problem components or functions can be selected for pruning. Building on existing work (Wang et al. 2024). To support the ranking of functions on the critical action path, we use triangular fuzzy numbers (TFNs) to calculate the importance of functions, addressing the uncertainty of experts' personal preferences and supervisors' decision-making. This allows us to prioritise functions with higher weights for pruning analysis, simplifying the solution paths of critical functions. The specific process is outlined as follows:

(1) Constructing a judgment matrix

When comparing the importance of any two elements, their importance ratio is quantified, expressing the relative importance of the two elements for evaluation on a scale ranging from 0.1 to 0.9. The scales of the TFNs and the definition of the elements in the matrix are shown in Table 2.

(2) Ranking functional importance based on functional value

Combined with the importance ranking method based on TFNs, the value of each functional unit constituting the Smart PSS is evaluated. The ranking method involves the following steps (specific steps detailed in Appendix A).

Normalise all functional units in the system based on TFNs and construct a complementary TFNs matrix, referring to Table 2. We construct the TFNS complementary matrix for the Smart Forklift Product-Service System as follows:

$$M = \begin{bmatrix} (0.50, 0.50, 0.50) & (0.20, 0.30, 0.40) & (0.50, 0.55, 0.60) & (0.65, 0.75, 0.80) & (0.20, 0.35, 0.40) \\ (0.60, 0.70, 0.80) & (0.50, 0.50, 0.50) & (0.55, 0.60, 0.70) & (0.60, 0.75, 0.80) & (0.20, 0.25, 0.35) \\ (0.40, 0.45, 0.50) & (0.30, 0.40, 0.45) & (0.50, 0.50, 0.50) & (0.65, 0.75, 0.80) & (0.30, 0.35, 0.45) \\ (0.50, 0.60, 0.65) & (0.25, 0.30, 0.40) & (0.30, 0.35, 0.45) & (0.50, 0.50, 0.50) & (0.45, 0.55, 0.60) \\ (0.60, 0.65, 0.75) & (0.30, 0.35, 0.45) & (0.25, 0.30, 0.40) & (0.30, 0.35, 0.45) & (0.50, 0.50, 0.50) \end{bmatrix}$$

Then we calculate the weighting vector of Fuzzy Ordered Weighted Averaging (FOWA) and the expected values of each element in the complementary matrix $\tilde{A}$ to derive the importance values of each functional unit. After normalisation and sorting, calculate the total weight values of each functional unit based on the judgments of all raters. Ultimately, rank all functional units of the target Smart PSS by importance, prioritising the pruning of functions with higher weights to simplify the solution path and address problems effectively.

Step 6: Functional Pruning and Pruning Problem Model Analysis.

The implementation of pruning involves a subjective analysis by designers following the resource search under the guidance of pruning rules. Pruning rules should be based on the compositional characteristics of the Smart PSS. In the functional pruning of the Smart PSS, identify pruning rules that meet requirements through specific analysis. Proceed to the next step if there is a new problem in the pruning model.

If the pruning process yields a pruned model with pruning problems, these issues may give rise to various types of design-related problems, including engineering contradiction problems, physical contradiction problems and matter-field problems. For different pruning scenarios, suitable tools such as the separation principle, matter-field analysis, standard solution methods and effects can be used to address the pruned system problems. Alternatively, heuristic solutions can be analysed using the effect library or the patent knowledge base. In this paper, the cross-domain search of knowledge from useful patent libraries is predominantly chosen as the approach to solving these problems.

3.3.2. *Knowledge search based on function similarity*

In Smart PSS design, designers need to access existing solutions and knowledge from the knowledge base and thoroughly evaluate function information and function semantic similarity from patented knowledge as a source of innovative knowledge for designing Smart PSS.

The similarity between requirement functions and patent knowledge sources in Smart PSS design primarily considers the degree of similarity between KGs. In this paper, we propose a ranking method that combines node similarity and node weights for the design knowledge graph that best meets the requirements. This method mainly includes the following aspects:

(1) Functional Similarity

In search modelling, to acquire a broader range of relevant cross-domain patent text information, we adopted the classical 'verb-noun' functional description method based on existing functional basic research (Pahl et al. 2007). This method aims to minimise limitations imposed by functional application domains, thereby enhancing opportunities for identifying more innovative solutions. Such an approach ensures that the retrieved knowledge is more comprehensive. Subsequently, we use a transformer-based BERT model to evaluate functional similarities, thereby identifying patent knowledge that is more suitable.

For phrase similarity calculations based on BERT (Devlin et al. 2018), we leverage the BertTokenizer and BertModel from the transformers library. The process begins with encoding the input phrases using BertTokenizer, that is, the continuous text data is converted into a discrete sequence of symbols. Assuming that the text is a string T, the tokenisation process can be represented as a function f maps T to a sequence of tokens $\{t_1, t_2, \ldots, t_n\}$, where $f(T) = \{t_1, t_2, \ldots, t_n\}$, where t_i is a token in the text.

The next step required is encoding, that is, converting the sequence of tokens into a numeric vector or tensor. Each token is first mapped to a unique integer index. If the vocabulary V contains all possible tokens, the mapping function g can be defined as $g(t) = i$, where $t \in$ V and i is the index of t. The indexed sequence is $\{g(t_1), g(t_2), \ldots, g(t_n)\}$. Then special markers are added, and special markers [CLS] and [SEP] are added to indicate the beginning and end of a sentence, or the separation between sentences. Suppose the sequence of tokens is $\{t_1, t_2, \ldots, t_n\}$, and the sequence after adding special tokens is $\{[CLS]t_1, t_2, \ldots, t_n, [SEP]\}$. Finally tensorization is performed. The indexed sequence is converted into tensor form, suitable for model processing. If there is a sequence $\{i_1, i_2, \ldots, i_m\}$, the tensorisation can be expressed as a vector $X = \{i_1, i_2, \ldots, i_m\}$. This transformation usually represents a one-dimensional indexed sequence as a two-dimensional tensor in order to handle batches (batch).

We calculate the dot product of the two vectors and the vector's paradigm (i.e. length) by taking two text strings, processed as described above, and the dot product divided by the product of the two vector's paradigms is the cosine similarity. This process is implemented using Python code (in Appendix B).

(2) Similarity Assessment of Knowledge Graphs

In order to accurately match candidate KGs from the knowledge base, the question about the similarity measure between the design problem and candidate patent knowledge is converted into a similarity measure between two directed relationship graphs. This means that the graph structure and node types of the two matching graphs should be as similar as possible.

The process involves calculating weights on the nodes for each node, performing similarity calculations and ranking all candidate KGs and design problems. The knowledge graph with the highest similarity value is then selected as the design knowledge source closest to the design requirements (details provided in Appendix C).

Step 7: Knowledge-Supported Scenario-Based Solving.

To effectively support the conceptual design of a Smart PSS, it is essential to extract the relevant understanding corresponding to a specific problem from a vast and intricate

Table 3. Requirements of Smart Forklift Product-Service Systems.

Serial number	Design requirements	Functional requirements
1	Req1: The system should transport the goods to the specified location.	S1: Transport
2	Req2: Power should be supplied to the system.	S2: Energy supply
3	Req3: It should be about securing the system and the operator.	S3: Protection
4	Req4: System performance should be monitored.	S4: Condition monitoring
5	Req5: The system should automatically arrive at the desired location.	S5: Positioning
6	Req6: Goods should be stored properly.	S6: Cargo storage
7	Req7: Basic customer operation/maintenance skills should be acquired through training.	S7: Training
8	Req8: There should be a customised aftermarket survey.	S8: After-sales survey

patent knowledge base and apply this knowledge effectively. This involves formalising and generalising the design problem and then matching it with design knowledge in the patent library to find design patents with similar characteristics.

4. Case studies

A company provides forklifts to its customers and engages in R&D, production and sales of product parts and components. It has undergone digital transformation in recent years. Confronting the impact of intelligent manufacturing, the company must establish an intelligent warehousing and logistics service system to enhance overall operational efficiency and intelligence. Prior to the introduction of smart forklifts, material turnover in the production workshop was frequent, with large and heavy materials. Manual handling incurred a significant workload and certain safety risks. To enable real-time coordination and integration of material control and management, a smart forklift service system is proposed. Through communication interconnection among different devices, management personnel can make effective decisions based on information to achieve intelligent warehousing goals. The proposed methods are implemented as follows:

Step 1: Define the System Requirements.

Using LLM, we provide the business processes, usage scenarios, and stakeholder requirements of the smart PSS to LLM, and extract the main design requirements of the system and the corresponding functional requirements, as shown in Table 3.

Step 2: Clarification of System Functions.

Based on the service effect provided by the Smart PSS for smart forklifts, the system's primary functions are clarified and decomposed, and the system function tree is shown in Figure 7.

Step 3: Clarification of System Boundaries.

The proposed smart forklift PSS comprises logistics handling, a transmission system and a corresponding control and management system. It relies on the smart forklift as the core equipment to perform a series of functions collectively, providing a reliable guarantee for

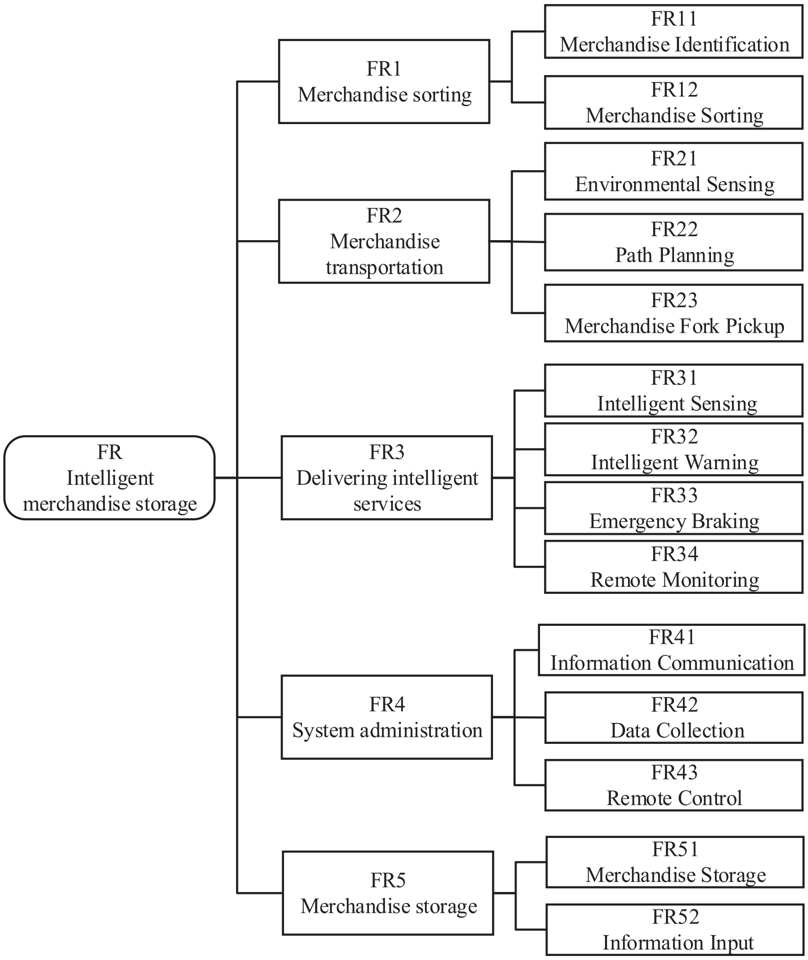

Figure 7. Function Tree of Smart Forklift Product-Service System.

the efficiency and accuracy of warehousing operations. As a complex organisation integrating multidisciplinary technologies, the smart forklift PSS primarily consists of three parts: a management platform, execution equipment and an intelligent service system, as shown in Figure 8.

Step 4: Multidimensional Functional Domain Modelling.
(1) Functional Domain Analysis of Physical Entities

In the physical entity domain, required functions include machine on/off, forklift operation, environment sensing, collision warning and material forking. After defining these functions, the conceptual structure of the physical entity module is further analysed as follows:

① Machine start/stop: An intelligent terminal or human-machine interface, along with energy input, initiates and halts machine operation.

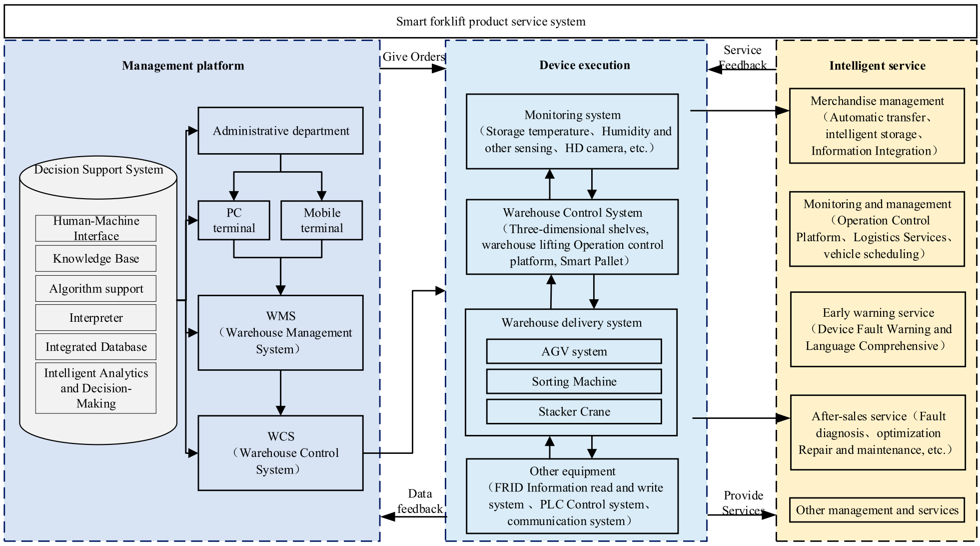

Figure 8. Smart Forklift Product-Service System Components.

② Machine movement: Forklift travel requires cooperation among the drive motor, wheels and rotary encoder.

③ Environmental sensing: This function requires collaboration between the obstacle detection system and its position detection system, including components such as a LIDAR system, infrared ranging sensor and mechanical collision sensing sensor.

④ Collision warning: Smart forklift trucks feature real-time task process feedback and emergency response, along with alarm functions, such as lights and voice alerts.

⑤ Material forking: The detecting mechanism assesses goods condition and signals the control module to operate the fork mechanism for material retrieval.

The physical entity module serves for material transportation management, primarily carried out by diverse execution equipment with a smart forklift at its core. Realising the function of this module entails not only considering the smart forklift but also monitoring storage conditions such as temperature, humidity and other environmental factors. This involves using sensors, cameras, intelligent terminals and other equipment.

(2) Software Functional Domain Analysis

The software within a smart forklift PSS encompasses controls such as Warehouse Control System (WCS) management for smart forklifts and other warehouse equipment, involving dynamic planning, monitoring, equipment communication docking, traffic control, diagnostics and more. Once elements of the software functional domain are identified, the conceptual scheme of the software modules of the smart forklift PSS can be schematised.

(3) Analysis of Service Functional Areas

In the service module of the system, the interaction between the user and the service module operates in both directions. Users can issue various control commands to the smart forklift via the mobile application, such as defining the operation area or adjusting the working route, for personalised service. Additionally, the forklift provides real-time feedback on its working status to the user, including information like the robot's working trajectory and the forklift's remaining power, enabling real-time status detection. Furthermore, the cloud platform provided by the supplier offers remote upgrade and management services for the application software.

Based on the above analysis, the initial conceptual structure of a smart forklift PSS is completed using the functional modelling, as shown in Figure 9.

Step 5: Determine the Pruning Object.

For the system functional modelling applying pruning, the priority determination analysis reverses the flow path in the model. The system defect flow includes:

Cargo forklift: The forklift struggles to align the cargo position's height accurately, resulting in low precision. This issue can lead to cargo instability and potential rollovers. Additionally, the forklift may fail to accurately lift cargo from specified positions, failing to meet design expectations. A schematic diagram illustrating the extraction of the defect flow tracking path is shown in Figure 10.

After identifying the critical conflict area, the function of this unfavourable path is further improved by sorting and filtering through function value calculation according to the method of determining the trimming object proposed in Section 3.3.1. Following investigation, analysis and evaluation by experts in relevant fields, functions of the smart forklift PSS are compared according to Table 4. A TFN complementary decision matrix is then formed as follows:

$$M=\begin{bmatrix} (0.50,0.50,0.50) & (0.20,0.30,0.40) & (0.50,0.55,0.60) & (0.65,0.75,0.80) & (0.20,0.35,0.40) \\ (0.60,0.70,0.80) & (0.50,0.50,0.50) & (0.55,0.60,0.70) & (0.60,0.75,0.80) & (0.20,0.25,0.35) \\ (0.40,0.45,0.50) & (0.30,0.40,0.45) & (0.50,0.50,0.50) & (0.65,0.75,0.80) & (0.30,0.35,0.45) \\ (0.50,0.60,0.65) & (0.25,0.30,0.40) & (0.30,0.35,0.45) & (0.50,0.50,0.50) & (0.45,0.55,0.60) \\ (0.60,0.65,0.75) & (0.30,0.35,0.45) & (0.25,0.30,0.40) & (0.30,0.35,0.45) & (0.50,0.50,0.50) \end{bmatrix}$$

The complementary matrix is processed to determine the importance of functions in the system by normalising all functional units, as shown in Table 4.

From the comprehensive analysis above, it is evident that forklift cargo handling holds the highest priority for pruning and is identified as the area requiring optimisation.

Step 6: Functional Pruning and Pruning Problem Model Analysis.

After the initial screening and analysis, suitable pruning rules are sought to refine the service system. Optional pruning rules are outlined in Table 5.

Step 7: Knowledge-Supported Scenario-Based Solving.

(1) When Rule 2 is selected for pruning, it is discovered that the forked goods fail to meet design expectations due to the inability to align the height of the goods' position accurately. Additionally, the low precision makes it impossible to fork goods from specified

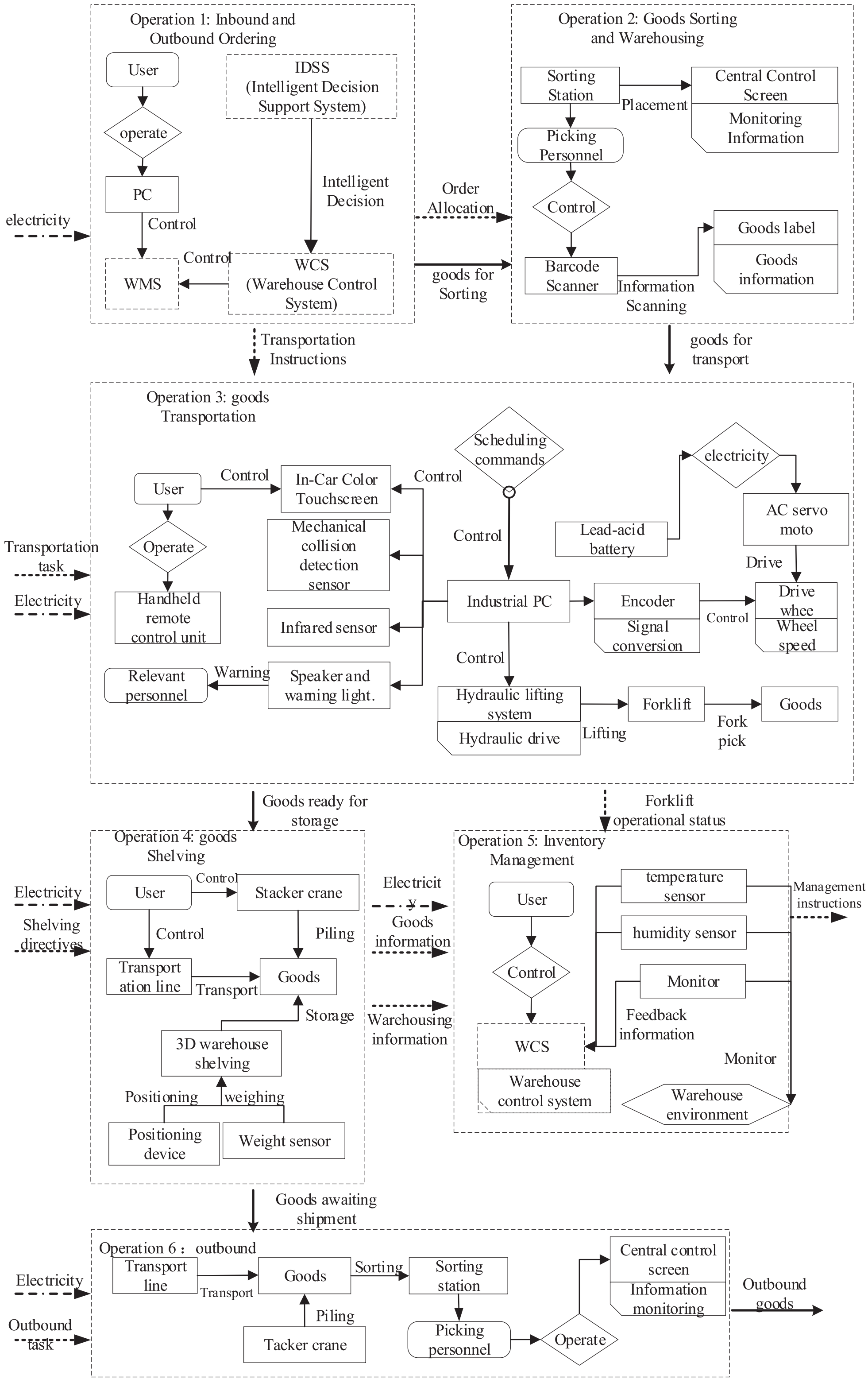

Figure 9. Initial Functional modelling of Smart Forklift Product-Service System.

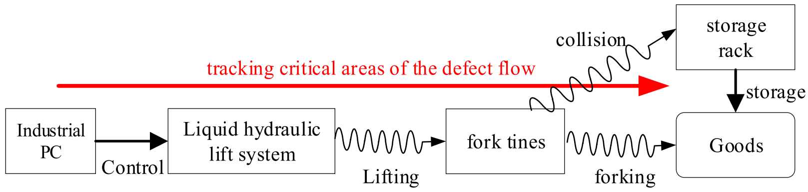

Figure 10. Defect Flow Tracing Path.

Table 4. Ranking the importance of system function.

Functionality	Significance	Pruning priority
Lifting forks	0.082	5
Containment	0.117	4
Forklift cargo	0.171	1
Collisions	0.123	3
Storage of goods	0.161	2

Table 5. Feasibility analysis of cropping rules.

Code number	Content of the rules	Feasibility analysis
Rule 1	Finding new technologies and principles to replace the subsystem's original technical solutions.	It can be used as an alternative rule.
Rule 2	Optimise target module parameters for self-satisfying functionality.	Improvements can be made by finding new technological solutions.
Rule 3	Delete the problem module directly.	The deletion of the gantry from the smart forklift is not chosen because, as indicated by the functional modelling, removing the gantry would prevent the successful lifting of goods.
Rule 4	Look for integration with subsystems within the system to improve its functional capabilities.	Analysis of the functional modelling reveals minimal interaction between this process and other subsystems, leading to the decision not to select this rule.

Table 6. Available design knowledge after screening.

Number	Patent name	Patent number	Functional similarity
1	A kind of alarm device for car anti-collision	CN212765951U	0.106
2	A kind of CNC machine tool spindle vibration monitoring system	CN109724686A	0.528
3	A vehicle distance measurement method, device and automobile	CN103148837B	0.605
4	Height measuring equipment and methods	CN115265384A	0.223

positions accurately. To optimise parameters of the target module for self-satisfaction, the critical function identified is 'detecting position'.

To enhance 'detection position', relevant topics are explored in the patent database. Representative patents are selected as design references based on design requirements, as shown in Figure 11. Since existing design knowledge may not fully satisfy all functional requirements, representative knowledge is filtered based on functional similarity and listed in Table 6. The design concept is then formulated through the knowledge fusion strategy proposed in this paper.

References are made using the distance measuring method, leading to the formation of an innovative solution, S2, for the forklift, as shown in Figure 12. In this solution, a distance

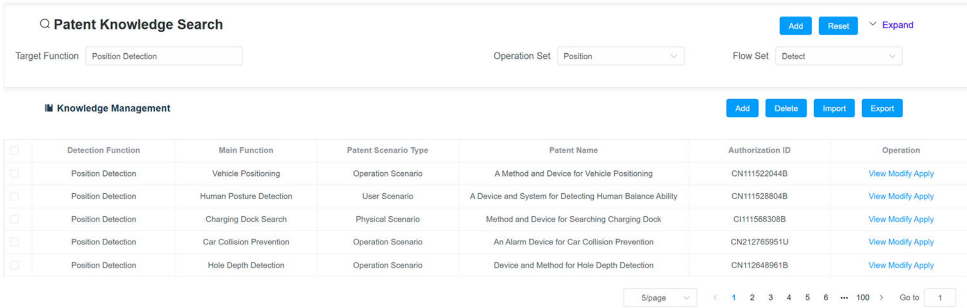

Figure 11. Design Knowledge Search Results.

detector is employed to detect the distance between the fork and the target, transmitting this distance to the control module for precise operation.

(2) When Rule 3 is selected for pruning, the objective is to find new technologies and principles to replace the original subsystem principles fundamentally. Using the search term 'mobile goods' in the patent knowledge base, the knowledge source found is for the magnetic drive clamping fibre disc clamping lifting device (CN109399493A) principle. This forms an innovation scheme, S3, as follows: A logistics handling fork comprising a fork and bearing plate. The bearing plate features permanent magnet plates on both sides and an electromagnetic plate in the middle. By changing the plate current direction, the electromagnetic plate pushes the permanent magnet plates to slide. This design enlarges the contact area of the load-bearing plate to prevent tilting or overturning of the pallet in case of inaccurate positioning or positioning errors. Additionally, a hydraulic device is used for buffering to protect wear parts from the impact of significant instantaneous pressure.

The smart forklift PSS, functioning as an integrated solution of products and services, can leverage FOS-based knowledge support for enhancing the service dimension of the system. Taking the intelligent monitoring service of the smart forklift PSS as an example, a corresponding search is conducted in the patent knowledge base. This search references knowledge sources, such as the patent for remote monitoring methods, systems, devices and mediums in pipeline production processes (CN117148775A). Based on sensors deployed at the worksite, real-time sensory information of the smart forklift's operation is acquired. This sensory information is transmitted to edge computing devices associated with the production site. These edge computing devices process this information in real-time and then transmit it to the cloud. In the cloud, work parameters are adjusted through remote control functions and the operating status parameters of the smart forklift are visually displayed. This process enables better provision of personalised services to relevant stakeholders.

Through the process outlined above, cross-domain design knowledge is integrated into the design of the smart forklift PSS. By conducting comparative analysis of design results, the final conceptual scheme for the smart forklift PSS is generated, as shown in Figure 13. Explanations for the relevant constituent elements are provided in Table 7.

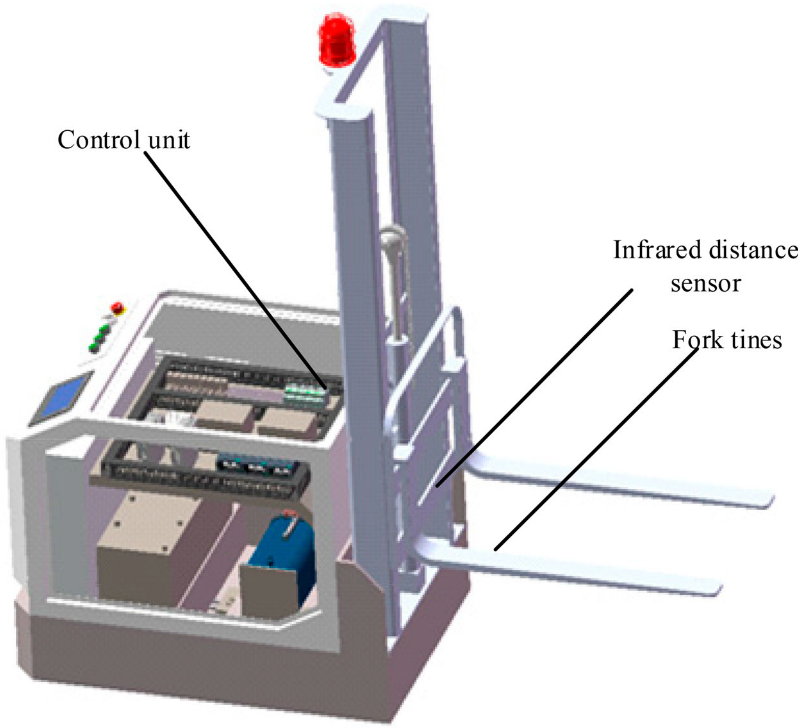

Figure 12. Programme Description of Distance Detection Function.

Table 7. Constituent elements of the overall programme of the Smart Forklift product-service system.

Constituent element	Description
Physical device	Smart forklift: vehicle body, vehicle controller, autonomous positioning and guidance system, motion control system, motor drive assembly, energy system, wireless communication system, etc. Other workshop facilities: three-dimensional shelves, video monitoring equipment, signal acquisition equipment, etc.
Software system	Smart forklift Dispatch Management System: manages hundreds of robot fleets, traffic control, implementation monitoring and collaborative scheduling to meet the demand for efficient operation of complex tasks in the production line.
Client service	Workshop operation scene control, data resource sharing service, employee and forklift guidance service, facility asset management and maintenance.
Device connection	DSRC, LTE-V, 4G/5G communication networks, Wi-Fi, RFID, CAN, Bluetooth.
Supporting equipment and systems	Robot business logic and action teaching system + manual operation remote control handle: supporting software system facilitating front-line workers to quickly get started with the application, enabling zero-threshold operation and maintenance of smart forklifts.
Customer value	Enhances work efficiency, saves costs and brings greater convenience to production and management.
Business model	Provide overall solutions: Establishes intelligent warehouses for enterprises to provide services integrating planning and design, product design, software development, on-site installation and after-sales service.

After the final functional modelling of the system is generated, real-time simulation is proposed for the entire operation process of the system, as shown in Figure 14.

The execution layer comprises the WCS and physical terminal equipment. The WCS receives scheduling instructions from the warehouse management system. It then sends

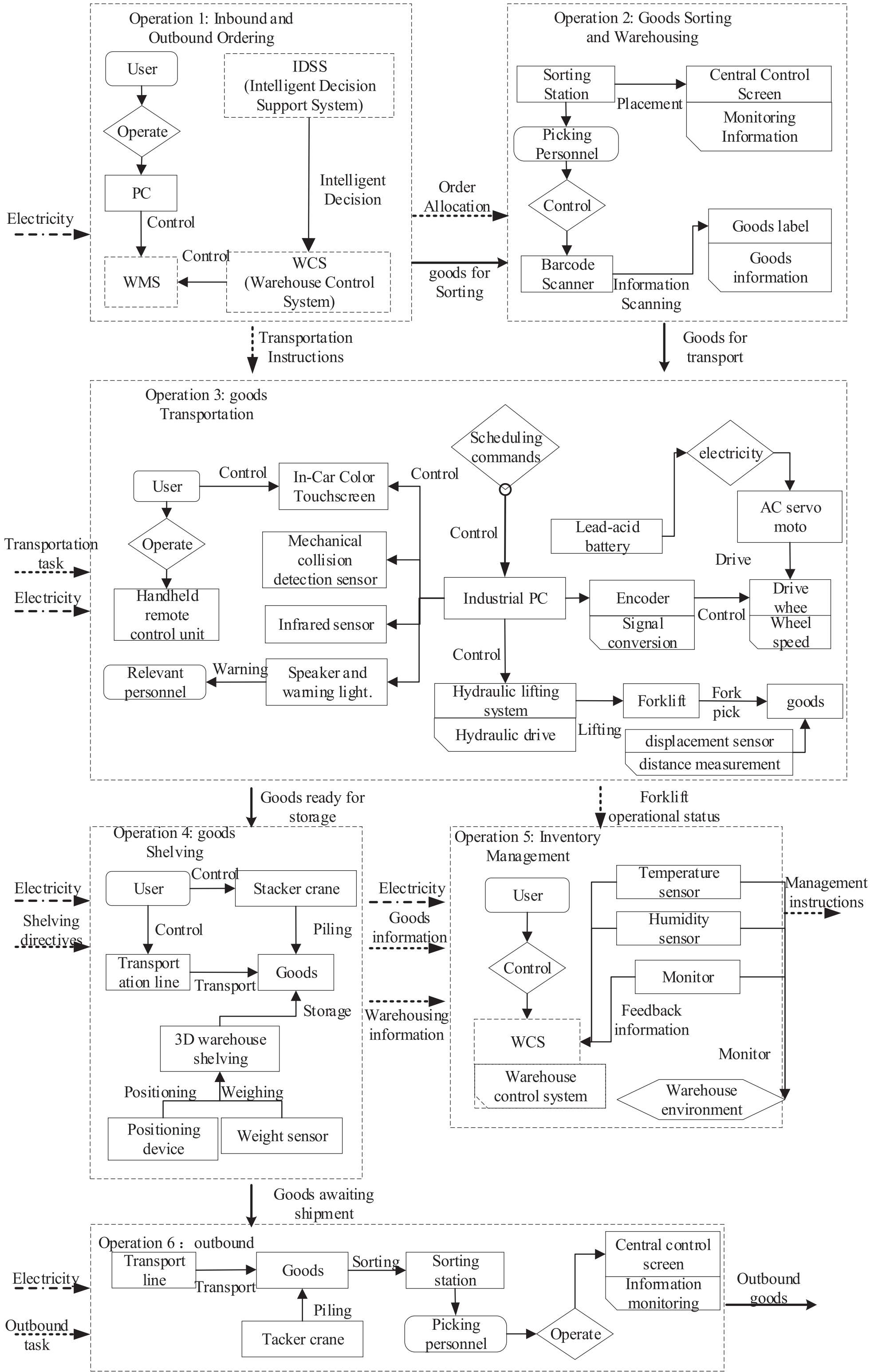

Figure 13. Conceptual Structure of the Smart Forklift Product-Service System.

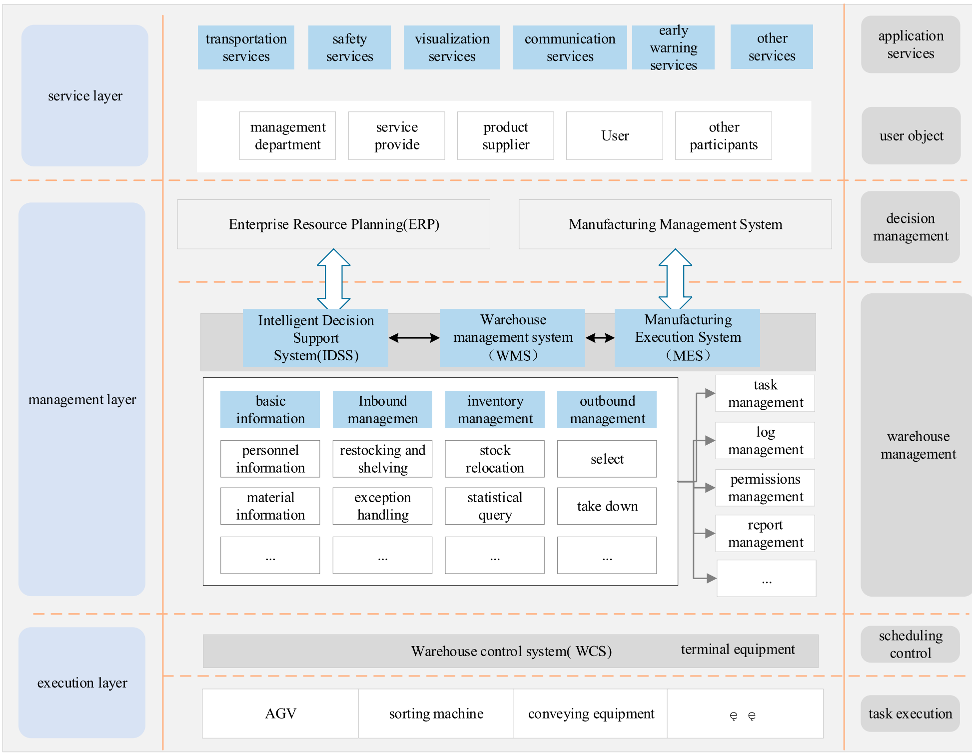

Figure 14. Overall Framework for the Smart Forklift Product-Service System.

control and execution instructions to the physical terminal equipment, which subsequently completes production activities, such as assembly, transportation and warehousing of goods.

The management level encompasses the warehouse management system, manufacturing execution system, etc., serving as the decision-making centre for warehouse management.

The service layer targets user objects, including management departments, service providers, drivers and users. It provides various services such as safety, communication, supervision and visualisation services tailored to different stakeholders.

5. Conclusion and further work

Based on the multifunctional coupling characteristics of smart PSS and the time-varying characteristics of the functions according to the user's needs, this study solves the functional coupling problem of smart PSS in the conceptual design process from the functional point of view. We propose a method for knowledge representation and reuse tailored to the design of Smart PSS, aided by FOS tools. Based on the characteristics of Smart PSS, this paper proposes the use of LLM to assist designers in capturing design requirements. Addressing the limitations of previous functional modelling methods in effectively representing the functional relationships of Smart PSS, the structural composition of Smart PSS is clarified, and a multidimensional functional domain-based modelling approach is introduced. By extending the FBS model, Smart PSS design knowledge representation is constructed. A function-oriented patent knowledge retrieval method is proposed, utilising a similarity calculation model that combines node and node-weight relationships in the knowledge graph, enabling the effective organisation of cross-domain design knowledge. Through knowledge search, the functional pruning and knowledge integration of Smart PSS are achieved, leading to its conceptual design. This approach overcomes the challenges in previous modelling methods related to multifunctional coupling in Smart PSS conceptual design and cross-domain knowledge transfer.

Compared to other current advanced modelling methods, such as the System Engineering framework or the Service Engineering Methodology, this paper places greater emphasis on functional integration and knowledge reuse, overcoming challenges related to multifunctional coupling in the conceptual design of Smart PSS and cross-domain knowledge transfer that existed in previous modelling methods. However, the current work may not have addressed key aspects such as lifecycle management, sustainability, and user experience. Additionally, since this paper is based on function-oriented patent knowledge search, its effectiveness may be limited in fields where patent information is scarce or not readily accessible.

Future research should strive to address these issues by continuously conducting empirical studies of the method in practical applications and incorporating aspects such as lifecycle management and sustainability into the framework. This would provide a more comprehensive solution for the conceptual design of Smart PSS, enhancing the practical applicability of the conceptual model and making it a valuable tool for designers and engineers working in the field of Smart PSS.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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CRediT authorship contribution statement

Chunlong Wu: Validation, Resources, Supervision, Funding acquisition, Writing – review & editing. Xingwang Wang: Conceptualisation, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. Tao Chen: Resources, Investigation, Writing – review& editing. Hao Li: Writing – review. Haoqi Wang: review.

Data availability statement

No data was used for the research described in the article.

References

Abdoli, Shiva, and Sami Kara. 2020. "A Modelling Framework to Support Design of Complex Engineering Systems in Early Design Stages." *Research in Engineering Design* 31 (1): 25–52.

Altshuller, G. S. 1999. *The Innovation Algorithm: TRIZ, Systematic Innovation and Technical Creativity*. Worcester: Technical Innovation Center, Inc.

Bo, Y., G. Changqing, W. Xingzu, Y. Kong, and Y. Xiaoling. 2014. "Conceptual Design of Product Structure Based on Unit Model." *Computer Integrated Manufacturing Systems* 20 (3): 471–485. (in Chinese)

Buede, D. M., and William D. Miller. 2024. *The Engineering Design of Systems: Models and Methods*. Hoboken, NJ: Wiley.

Caldwell, B. W., and G. M. Mocko. 2012. "Validation of Function Pruning Rules Through Similarity at Three Levels of Abstraction." *Journal of Mechanical Design* 134 (3): 123–135.

Chang, D., Z. Gu, F. Li, and R. Jiang. 2019. "A User-Centric Innovative Product-Service System Development Approach: A Case Study on Medication Management for the Elderly." *Advanced Engineering Informatics* 42:100979.

Chao, M.H., and Amy J.C. Trappey. 2024. "Design of Product Configuration Systems Supporting Customised Product Cost Estimation Using Object-Oriented Ontology Framework." *Journal of Engineering Design* 35 (6): 709–742.

Chen, B., and Y. Xie. 2017. "Functional Knowledge Integration of the Design Process." *Science China Technological Sciences* 60 (2): 209–218.

Choi, S. C., D. W. Kang, J. Y. Lim, and K. Kim. 2012. "A Fact-Oriented Ontological Approach to SAO-Based Function Modelling of Patents for Implementing Function-Based Technology Database." *Expert Systems with Applications* 39 (10): 312–317.

Cong, J., C. H. Chen, P. Zheng, X. Li, and Z. Wang. 2020. "A Holistic Relook at Engineering Design Methodologies for Smart Product Service Systems Development." *Journal of Cleaner Production* 272:122737.

Cui, H., L. Gong, S. Huang, J. Leng, P. Zheng, and Y. Yan. 2024. "Cognitive Intelligence-Enabled Requirements Elicitation for Design Optimisation of Smart Product-Service System." *Journal of Engineering Design*, 1–25.

Devlin, J., M. Chang, K. Lee, and K. Toutanova. 2018. "Bert: Pre-training of Deep Bidirectional Transformers for Language Understanding." arXiv preprint arXiv:1810.04805.

Domb, E., and T. J. Kling. 2006. "How to Reduce Cost in Product and Process Using TRIZ." *TRIZ Journal*.

Dong, Yafan, Runhua Tan, Peng Zhang, Wei Liu, and Ruiqin Wang. 2019. "Research on Product Redesign Process Based on Functional Analysis." In *International Conference on Mechanical Design*, Vol. 77, 125–138. Singapore: Springer.

Eisenbart, B., K. Gericke, and L. T. M. Blessing. 2017. "Taking a Look at the Utilization of Function Models in Interdisciplinary Design: Insights from Ten Engineering Companies." *Research in Engineering Design* 28 (3): 299–331.

Fang, C., W. Liu, H. Lin, Y. Qi, X. Tian, Y. Huang, and S. J. Wang. 2024. "A User-Centred Collective System Design Approach for Smart Product-Service Systems: A Case Study on Fitness Product Design." *The Design Journal* 27 (3): 410–432.

Fantoni, G., R. Apreda, F. Dell'Orletta, and M. Monge. 2013. "Automatic Extraction of Function-Behavior-State Information from Patents." *Advanced Engineering Informatics* 27 (3): 317–334.

Gill, A. S., J. D. Summers, and C. Sen. 2019. "Impact of Chaining Method and Level of Completion on Accuracy of Function Structure-Based Market Price Prediction Models." *Journal of Computing and Information Science in Engineering* 19 (4): 041018.

Gui, F., and Y. Chen. 2021. "A Scenario-Integrated Approach for Functional Design of Smart Systems." *Artificial Intelligence for Engineering Design, Analysis and Manufacturing* 35 (2): 165–179.

Hagen, S., F. Kammler, and O. Thomas. 2018. "Adapting Product-Service System Methods for the Digital Era: Requirements for Smart PSS Engineering." Customization 4.0: proceedings of the 9th world mass customization & personalization conference (MCPC 2017), aachen, Germany, November 20th–21st, 2017. Springer International Publishing.

Hara, T., T. Arai, and Y. Shimomura. 2009. "A CAD System for Service Innovation: Integrated Representation of Function, Service Activity, and Product Behaviour." *Journal of Engineering Design* 20 (4): 367–388.

Jiang, S., H. Jie, K. L. Wood, and J. Luo. 2021. "Data-Driven Design-by-Analogy: State-of-the Art and Future Directions." *Journal of Mechanical Design* 144 (2): 020801.

Kang, Y., and D. Tang. 2014. "A Matrix-Based Solution Method for Multifunctional Product Conceptualization Scheme." *Computer Integrated Manufacturing Systems* 20 (12): 2915–2925.

Komoto, H., and T. Tomiyama. 2012. "A Framework for Computer-Aided Conceptual Design and its Application to System Architecting of Mechatronics Products." *Computer-Aided Design* 44 (10): 931–946.

Lee, C. H., C. H. Chen, and A. J. C. Trappey. 2019. "A Structural Service Innovation Approach for Designing Innovative Product-Service Systems: A Case Study of Intelligent Beauty Service." *Advanced Engineering Informatics* 40: 154–167.

Li, X., Chun-Hsien Chen, Pai Zheng, Zuoxu Wang, Zuhua Jiang, and Zhixing Jiang. 2020. "A Knowledge Graph-Aided Concept–Knowledge Approach for Evolutionary Smart Product–Service System Development." Journal of Mechanical Design 142 (10): 101403.

Litvin, S. 2005. "New TRIZ-Based Tool-Function-Oriented Search (FOS)." *The TRIZ Journal* 14:505–509.

Mirshafiee, N., J. Han, and S. Ahmed-Kristensen. 2024. "The DHSmart Model for Smart Product-Service System (Smart PSS): Dynamic, Data-Driven, Human-Centred." *Proceedings of the Design Society* 4: 2149–2158.

Mokhtarian, H., E. Coatanéa, and H. Paris. 2017. "Function Modelling Combined with Physics-Based Reasoning for Assessing Design Options and Supporting Innovative Ideation." *Artificial Intelligence for Engineering Design, Analysis, and Manufacturing* 31 (4): 476–500.

Montecchi, T., and D. Russo. 2015. "FBOS: Function/Behaviour–Oriented Search." *Procedia Engineering* 131: 140–149.

Pahl, G., W. Beitz, J. Feldhusen, and K. H. Grote. 2007. *Engineering Design: A Systematic Approach*. London: Springer-Verlag.

Russo, D. 2014. "Function-Based Patent Search: Achievements and Open Problems." *International Journal of Product Development* 19:060036.

Siddharth, L., Lucienne T. M. Blessing , Kristin L. Wood, and jianxi Luo. 2022. "Engineering Knowledge Graph from Patent Database." *American Society of Mechanical Engineers Digital Collection* 22 (2): 021008.

Singhal, A. 2012. "Introducing the Knowledge Graph: Things, Not Strings." May 16, 2012. https://blog.google/products/search/introducing-knowledge-graph-things-not/.

Valencia, A. 2017. *An Exploration of Innovative Product-Service System Design: Guidelines and Insights for Design Management*. Delft: Delft University of Technology.

Wang, Ning, Shan Ren, Yang Liu, Miying Yang, Jin Wang, and Donald Huisingh. 2024. "An Active Preventive Maintenance Approach of Complex Equipment Based on a Novel Product-Service System Operation Mode - ScienceDirect." *Journal of Cleaner Production* 277: 123365.

Wang, Boheng, Haoyu Zuo, Zebin Cai, Yuan Yin, Peter Childs, Lingyun Sun, and Liuqing Chen 2023. "A Task-Decomposed AI-Aided Approach for Generative Conceptual Design." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*. ASME, 87349: V006T06A009.

Wu, C. L., T. Chen, Z. Li, and W. Liu. 2021. "A Function-Oriented Optimizing Approach for Smart Product-Service Systems at the Conceptual Design Stage: A Perspective from the Digital Twin Framework." *Journal of Cleaner Production* 297: 126597.

Wu, Chunlong, Tianming Zhu, Peng Zhang, Jianguang SUN, and Runhua TAN. 2020. "Conceptual Scheme Construction of Smart Product Service System Based on Functional Model and Hierarchical Analysis." *China Mechanical Engineering* 31 (7): 853–864. 870. (in Chinese).

Xiang, H., W. Li, C. Li, and H. Wang. 2024. "An Evolutionary Graph of Product Service System Research Topics Based on Bibliometric Analysis and Topic Modelling." *Journal of Engineering Design* 35 (2): 131–158.

Xufen, J. 2021. "An Intelligent, Fast Return Back to Charging Seat Method for Sweeping Robot: CN111543897B."

Yan, X., Y. Li, J. Chen, W. Li, and Y. Xiong. 2017. "A Method of Implementing Formalized Multidisciplinary Collaboration in Product Conceptual Design Process." *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science* 231 (18): 3342–3357. https://doi.org/10.1177/0954406216646399.

Yu, Fei, Runhua Tan, Guozhong Cao, and Ping Jiang 2013. "Component Tailoring Prioritization Study Based on System Functional Modelling." *Computer Integrated Manufacturing Systems* 19 (2): 338–347. (in Chinese)

Zhang, H. 2010. *"Research on Product Patent Knowledge Acquisition and Its Method to Assist Product Innovation."* Hangzhou: Zhejiang University.

Zhang, Y., G. Jacobs, W. Yu, G. Hoepfner, and J. Berroth. 2023. "A System Modeling Process Based on SysML to Support Data Consistency Across System Requirement, Function, and Solution Model Layers." *Journal of Engineering Design* 34 (9): 674–690.

Zhang, Z., L. Liu, W. Wei, F. Tao, T. Li, and A. Liu. 2017. "A Systematic Function Recommendation Process for Data-Driven Product and Service Design." *Journal of Mechanical Design* 139 (11): 111404.

Zheng, P., T. J. Lin, C. H. Chen, and Xun Xu. 2018. "A Systematic Design Approach for Service Innovation of Smart Product-Service Systems." *Journal of Cleaner Production* 201 (1–1166): 657–667.

Zheng, Pai, Zuoxu Wang, Chun-Hsien Chen, Li Pheng Khoo. 2019. "A Survey of Smart Product-Service Systems: Key Aspects, Challenges and Future Perspectives." *Advanced Engineering Informatics* 42: 100973.

Appendices

Appendix 1: Methodology for ranking the importance of TFNs

(1) Normalisation of all functional units in the system based on TFNs

The Smart PSS, consisting of n functional units, can be represented by a complementary TFNs matrix as $\tilde{A} = (\tilde{a}_{ij})_{n\times n}$ where $\tilde{a}_{ij} = (a^L_{ij}, a^M_{ij}, a^N_{ij})$ concerning Table 2. The value of a_{ij} can be determined by comparing the importance of functional units i and j. Each a_{ij} is represented as a triad (l, M, U), where: The lower bound l represents the lowest relative importance that the expert believes function i has for function j in the most pessimistic scenario. The median value M represents the most likely relative importance value. The upper bound U represents the highest relative importance of i with respect to j in the most optimistic scenario for the expert.

(2) Combine Eq. (A1) and Eq. (A2) to compute the weighting vector of the FOWA.

The fuzzy semantic quantisation operator Q is determined by its upper and lower ends, denoted as Q (a, b), where a denotes the lower end and b denotes the upper end. To correspond to the fuzzy semantic quantisation criteria such as 'most', 'at least half' and 'as many as possible', the parameters in the operator Q are chosen as follows: For 'most', the parameters are (a, b) = (0.3, 0.8). For 'at least half', the parameters are (a, b) = (0, 0.5). For 'as many as possible', the parameters are (a, b) = (0.5, 1). Using such quantifier operators in decision-making or assessment processes helps translate subjective judgments into operational values for further analysis and comparison. By choosing appropriate values for a and b, decision-makers can reflect their risk preferences. For instance, choosing (0.3, 0.8) suggests that most cases should fall within this interval, representing a balance between being neither too conservative nor too aggressive, thus reflecting the expected or targeted situations in the majority of cases.

$$\omega_j = Q\left(\frac{j}{n}\right) - Q\left(\frac{j-1}{n}\right), \quad 1 \le j \le n \tag{A1}$$

$$Q(r) = \begin{cases} 0, & r < a \\ \dfrac{r-a}{b-a}, & a \le r \le b \\ 1, & r > b \end{cases} \tag{A2}$$

(3) Calculate the expected value in the complementary matrix $\tilde{A}$ using Eq. (A3).

$$\tilde{d}_i^{(\lambda)} = \frac{1}{2}[(1-\lambda)a_{ij}^L + a_{ij}^M + a_{ij}^N], \quad 1 \le i \le n, \ 1 \le j \le n, \ 0 \le \lambda \le 1 \tag{A3}$$

In the formula, the average growth amount of 0.5 is chosen as the normal preference. Then, the a_{ij} values are ranked according to their expected values, with the a_{ij} having the highest expected value ranked first in each row of the matrix. The ranking results are denoted as $\tilde{b}_{ik}$.

(4) Calculate the importance value of each functional unit by Eq. (A4) using the FOWA operator.

$$\tilde{d}_i = F(\tilde{a}_{i1}, \ldots \tilde{a}_{in}) = \omega_1 \otimes \tilde{b}_{i1} \oplus \omega_2 \otimes \tilde{b}_{i2} \oplus \ldots \omega_n \otimes \tilde{b}_{in} = \left[\sum_{i=1}^{n} \omega_k b_{ik}^L, \sum_{i=1}^{n} \omega_k b_{ik}^M, \sum_{i=1}^{n} \omega_k b_{ik}^U\right] \tag{A4}$$

(5) Calculate the expected value of each TFN through Eq. (A5) with the coefficient λ , 0.5.

$$\tilde{d}_i^{(\lambda)} = \frac{1}{2}[(1-\lambda)d_i^L + d_i^M + \lambda d_i^U] \tag{A5}$$

The expectation values are then further normalised as the final ranking weight using Eq. (A6) as an indication of the importance of each functional element.

$$\omega_i = \tilde{d}_i^{(\lambda)} / \sum_{j=1}^{n} \tilde{d}_i^{(\lambda)}, \quad 1 \le i \le n \tag{A6}$$

(6) Calculate the total weight value of each functional unit based on the judgment of all raters.

If m raters are recruited, the total weight value is calculated by Eq. (A7), where the weight for each rater in the group is denoted as

$$\omega_{Oi} = \sum_{j=1}^{m} \omega_{ji} v_j, \sum_{j=1}^{m} v_j = 1 \tag{A7}$$

Appendix 2: Similarity calculation code implementation

```
# Python code implementation of BERT-based phrase similarity computation
from transformers import BertTokenizer, BertModel
import torch
import numpy as np
# Initializing the BERT Model and Splitter
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')
# sample phrase
text1 = " "
text2 = " "
# Phrase encoding and embedding vector acquisition
encoded_input1 = tokenizer(text1,return_tensors = 'pt', add_special_tokens = True)
encoded_input2 = tokenizer(text2,return_tensors = 'pt', add_special_tokens = True)
with torch.no_grad():
    output1 = model(**encoded_input1)
    output2 = model(**encoded_input2)
vector1 = output1.last_hidden_state[:, 0, :].numpy()
vector2 = output2.last_hidden_state[:, 0, :].numpy()
#Calculate cosine similarity
cos_sim = np.dot(vector1,vector2.T)/(np.linalg.norm(vector1)* np.linalg.norm(vector2))
print("cosine similarity:", cos_sim[0][0])
```

Appendix 3: Knowledge graph similarity calculation method

The weight of each node is calculated as follows:

$$\omega_{i,j} = \left(1 + \frac{E_{i,j} - 1}{S_i}\right)\left(\frac{m - i + 1}{m}\right) \tag{A8}$$

In Equation (A8), $i = 1,2,\ldots m$, $\omega_{i,j}$ denotes the weight of the jth node located on the ith layer, $E_{i,j}$ denotes the number of edges starting from that n ode and S_i denotes the total number of edges of all nodes on the ith layer.

The similarity between the design problem graph and the kth candidate knowledge graph is as follows:

$$\mathrm{sim}(F_1, F_2) = \sum_{i=1} \sum_{j=1}^{n} \omega_{\mathrm{i,j}} \mathrm{sim}(N_{i,j}^{k}) \tag{A9}$$

where $N_{i,j}^{k}$ denotes the j node's similarity at level i, $\omega_{i,j}$ denotes the node's weight.